50 mm (1.97 in) bore × 30 mm (1.18 in) rod Cylinder Force Data

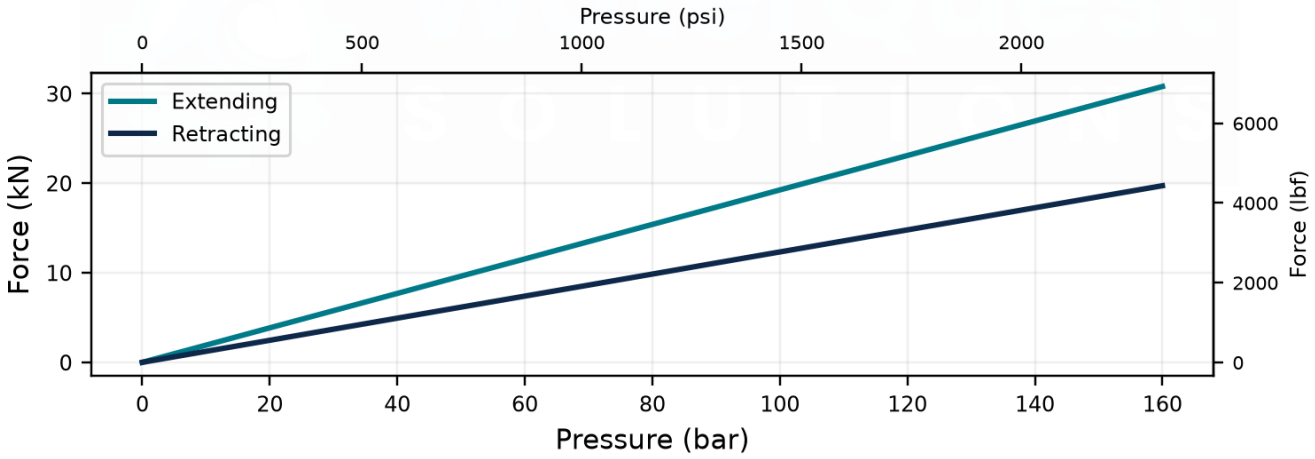
This geometry has a retraction area equal to 64.0% of its full piston area. Compare push and pull force at the same pressure, then check stroke, buckling, mounting and return-side pressure for the actual load.

Complete force output data across all operating pressures from 0 to 160 bar (0 to 2,321 psi). All values calculated with 0.98 efficiency factor.

At a glance

Parameter / value	Parameter / value
Piston Area 19.63 cm² (3.043 in²)	Ring Area 12.57 cm² (1.948 in²)
Max Extending Force 30.78 kN (6,920 lbf)	Max Retracting Force 19.71 kN (4,431 lbf)

Push and pull force



Plotted from the webpage force table, including its stated 0.98 efficiency factor.

Selection basis

Match the selected configuration to the working water, pressure, flow, temperature and duty. Maximum ratings and separate curve points do not establish one simultaneous operating point. Confirm installation and supplied configuration with WaterQuest Solutions.

Technical data

Specifications and selection data

Pressure	Force Extending (kN)	Force Retracting (kN)
0 bar (0 psi)	0.00 kN (0 lbf)	0.00 kN (0 lbf)
10 bar (145 psi)	1.92 kN (432 lbf)	1.23 kN (277 lbf)
20 bar (290 psi)	3.85 kN (866 lbf)	2.46 kN (553 lbf)
30 bar (435 psi)	5.77 kN (1,297 lbf)	3.70 kN (832 lbf)
40 bar (580 psi)	7.69 kN (1,729 lbf)	4.93 kN (1,108 lbf)
50 bar (725 psi)	9.62 kN (2,163 lbf)	6.16 kN (1,385 lbf)
60 bar (870 psi)	11.54 kN (2,594 lbf)	7.39 kN (1,661 lbf)
70 bar (1,015 psi)	13.47 kN (3,028 lbf)	8.62 kN (1,938 lbf)
80 bar (1,160 psi)	15.39 kN (3,460 lbf)	9.85 kN (2,214 lbf)
90 bar (1,305 psi)	17.31 kN (3,891 lbf)	11.09 kN (2,493 lbf)
100 bar (1,450 psi)	19.24 kN (4,325 lbf)	12.32 kN (2,770 lbf)
110 bar (1,595 psi)	21.16 kN (4,757 lbf)	13.55 kN (3,046 lbf)
120 bar (1,740 psi)	23.08 kN (5,189 lbf)	14.78 kN (3,323 lbf)
130 bar (1,885 psi)	25.01 kN (5,622 lbf)	16.01 kN (3,599 lbf)
140 bar (2,031 psi)	26.93 kN (6,054 lbf)	17.25 kN (3,878 lbf)
150 bar (2,176 psi)	28.86 kN (6,488 lbf)	18.48 kN (4,154 lbf)
160 bar (2,321 psi)	30.78 kN (6,920 lbf)	19.71 kN (4,431 lbf)